University of Prishtina “Hasan PRISHTINA”

Faculty of Electrical and Computer Engineering
Master in Computer and Software Engineering, 2025-2026
Course: Data Preparation and Visualization

Week VII – Anomaly Detection II

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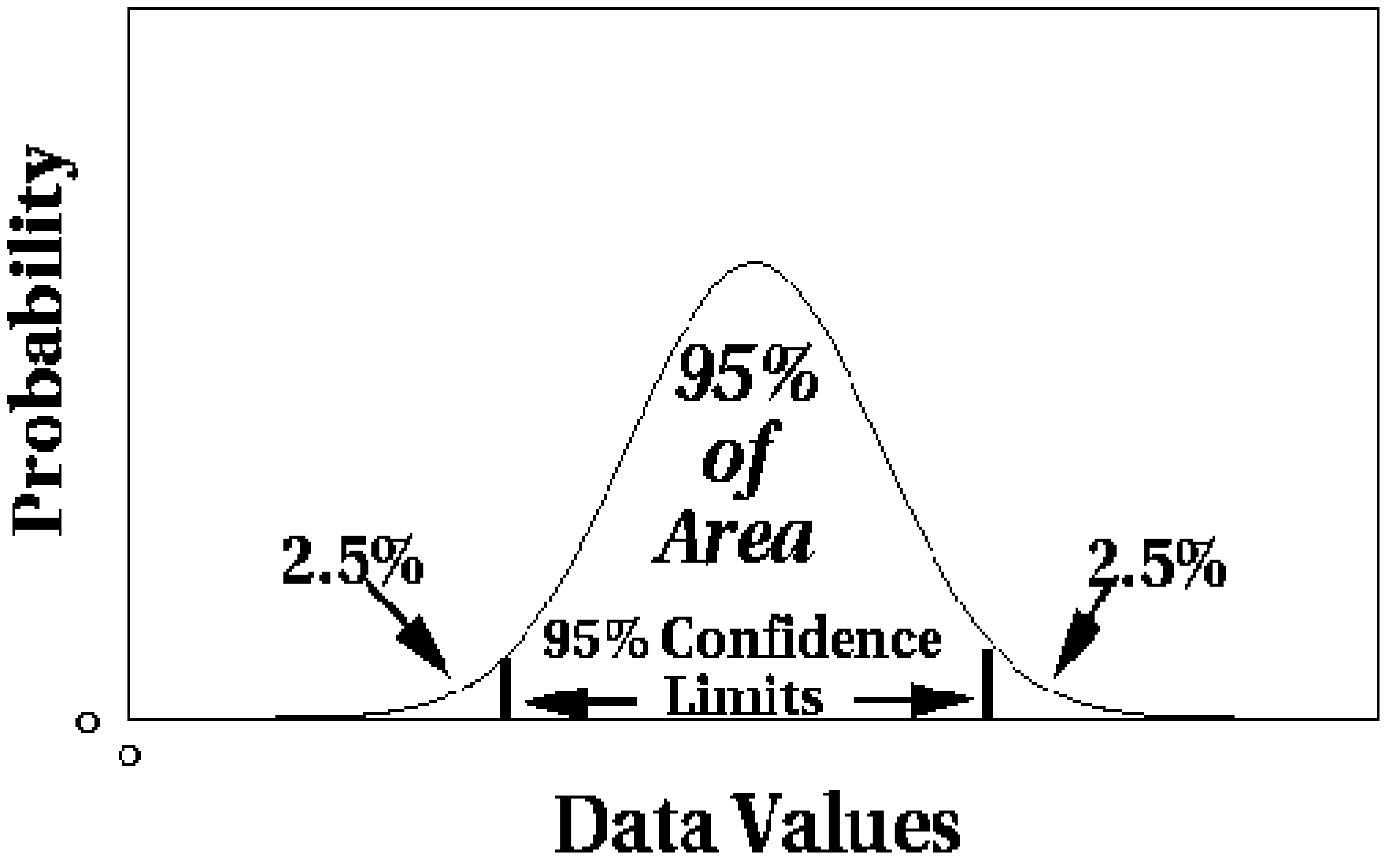
Today discussions

➢ **Which are techniques for anomaly detection?**

- Statistical;
- Proximity-based;
- Density-based;
- Clustering-based.

Statistical outlier detection

- Univariate Gaussian distribution:
  - Outlier defined by z-score > threshold.



Calculation of z-score

- The **z-score** is a statistical measure that describes **how many standard deviations** a data point is from the **mean of a dataset**;
- It is a way to standardize data and is often used to **identify outliers** or evaluate the relative standing of a value within a distribution;
- **Calculation of Z-score**:
  - The z-score of a data point x is calculated using the formula:

$\boldsymbol{z = x - \mu / \sigma}$

where: x=the data point;

$\boldsymbol{\mu}$ = the mean of the dataset;

$\boldsymbol{\sigma}$ = the standard deviation of the dataset

z = 0: The data point is exactly at the mean;

Positive z-score: The data point is above the mean;

Negative z-score: The data point is below the mean.

Typical Thresholds for Z-scores

- In practice, thresholds for z-scores are often set to identify outliers or unusual data points:

  - *$|z| > 2$: Data points are generally considered moderately unusual (about 95% of data falls within this range in a normal distribution).*
  - *$|z| > 3$: Data points are typically considered outliers (about 99.7% of data falls within this range in a normal distribution).*

  - $|z| \leq 2 \rightarrow$ ***normal***
  - $2 < |z| \leq 3 \rightarrow$ ***somewhat unusual***
  - $|z| > 3 \rightarrow$ ***very unusual / outlier***

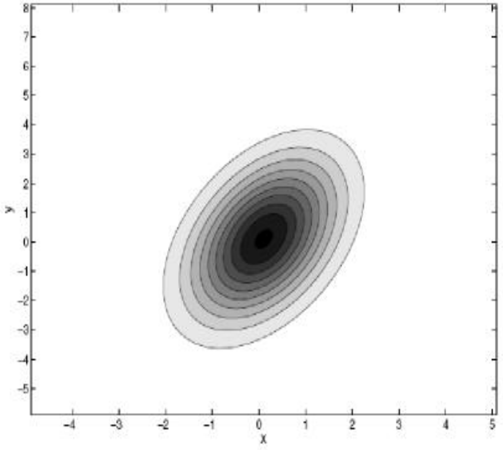
Is There a Static Threshold for Z-scores?

- A z-score threshold can be set as a static value (e.g., $|z| > 2$ or $|z| > 3$) for identifying outliers in data following a normal distribution;

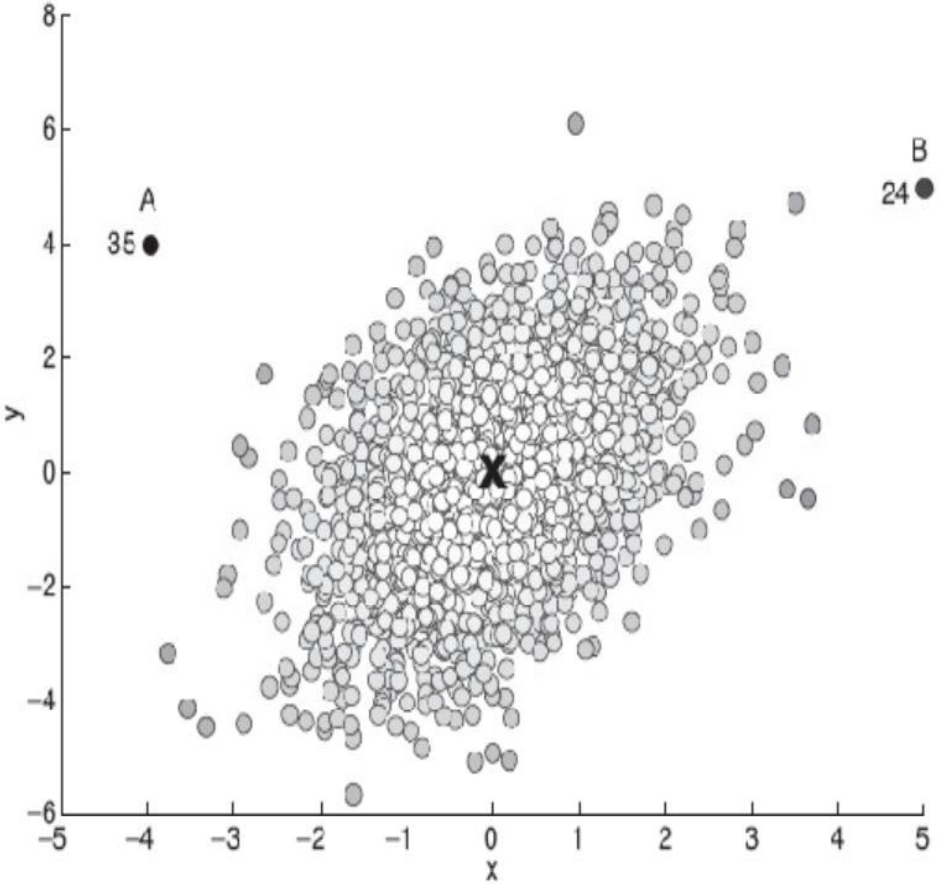
- However:
  - **Choice of Threshold**: The choice of z-score threshold can vary depending on the application, the desired sensitivity, and the nature of the data;
  - **Distribution Assumptions**: Z-score analysis assumes data is normally distributed. If the data distribution deviates significantly from normality, z-scores may not be as reliable, and other methods, such as percentiles, may be more appropriate.

Statistical anomaly detection

- Multivariate Gaussian distribution:
  - Outlier defined by Mahala Nobis distance > threshold



	Distance	
	Euclidean	Mahalanobis
A	5.7	35
B	7.1	24



Grubbs' test

- **Detect outliers** in univariate data;
- Assume data comes **from normal distribution**;
- Detects **one outlier at a time**, remove the **outlier**, and **repeat**:
  - H_0: There is no outlier in data;
  - H_A: There is at least one outlier
- Grubbs' test statistic:

$$G = \frac{\max\left|X - \overline{X}\right|}{s}$$

Xi is the suspect value.

$\bar{X}$ is the sample mean.

s is the sample

- Reject H_0 if:

$$G > \frac{(N-1)}{\sqrt{N}} \sqrt{\frac{t^2_{(\alpha/N, N-2)}}{N-2+t^2_{(\alpha/N, N-2)}}}$$

n is the sample size.

t is the critical value from the t-distribution for a two-tailed test at the chosen significance level.

Likelihood approach

- **Assume** the dataset **D contains samples** from a mixture of **two probability distributions**:
  - M (majority distribution);
  - A (anomalous distribution).
- **General approach:**
  - Initially, assume all the data points belong to M;
  - Let $L_t(D)$ be the log likelihood of D at time t;
  - For each point x_t that belongs to M, move it to A:
    - Let $L_{t+1}(D)$ be the new log likelihood;
    - Compute the difference, $\Delta = L_t(D) - L_{t+1}(D)$;
    - If $\Delta > c$ (some threshold), then x_t is declared as an anomaly and moved permanently from M to A.

Likelihood approach

- Data distribution, $D = (1 - \lambda) M + \lambda A$;
- M is a probability distribution estimated from data:
  - Can be based on any modelling method (***Naïve Bayes, maximum entropy, etc***)
- A is initially assumed to be uniform distribution;
- Likelihood at time t:

$$L_t(D) = \prod_{i=1}^{N} P_D(x_i) = \left((1-\lambda)^{|M_t|} \prod_{x_i \in M_t} P_{M_t}(x_i) \right) \left(\lambda^{|A_t|} \prod_{x_i \in A_t} P_{A_t}(x_i) \right)$$

$$LL_t(D) = |M_t| \log(1-\lambda) + \sum_{x_i \in M_t} \log P_{M_t}(x_i) + |A_t| \log \lambda + \sum_{x_i \in A_t} \log P_{A_t}(x_i)$$

Statistical outlier detection

- **Pros:**
  - Statistical tests are well-understood and well validated;
  - Quantitative measure of degree to which object is an outlier;
- **Cons:**
  - Data may be hard to model parametrically:
    - multiple modes;
    - variable density.
  - In **high dimensions**, data may be insufficient to **estimate true distribution**.

Proximity-based outlier detection

- **Outliers** are objects far away from other objects;

- **Common approach**:
  - **Outlier score** is distance **to k^t** nearest neighbour;
  - **Score sensitive** to choose of **k**.

Proximity-based outlier detection

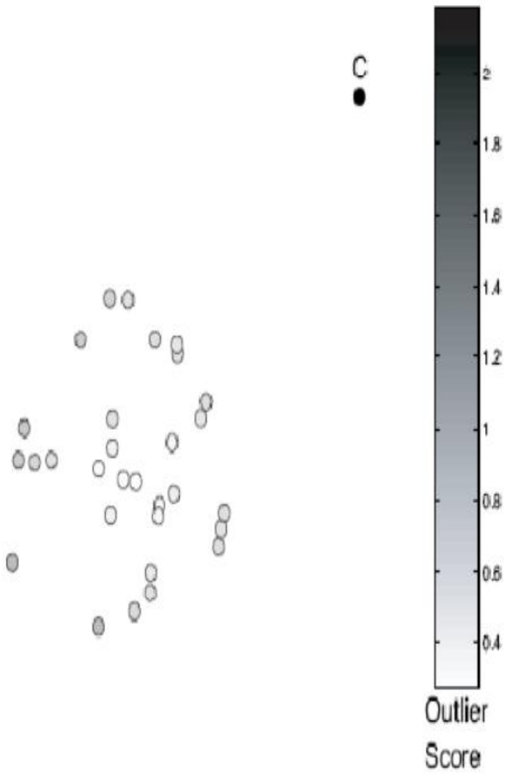


Figure 10.4. Outlier score based on the distance to fifth nearest neighbor.

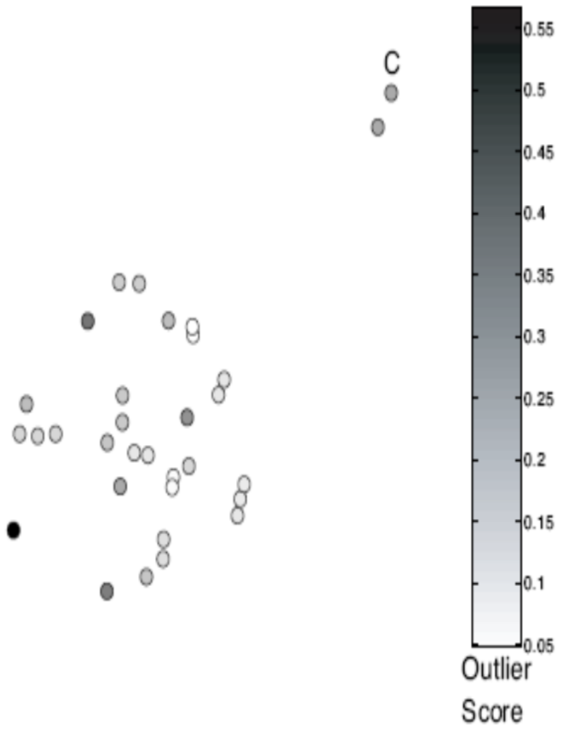


Figure 10.5. Outlier score based on the distance to the first nearest neighbor. Nearby outliers have low outlier scores.

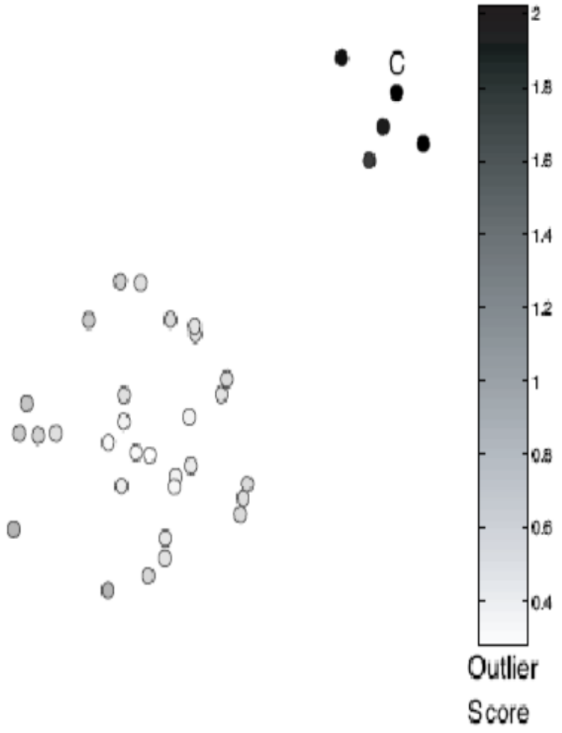


Figure 10.6. Outlier score based on distance to the fifth nearest neighbor. A small cluster becomes an outlier.

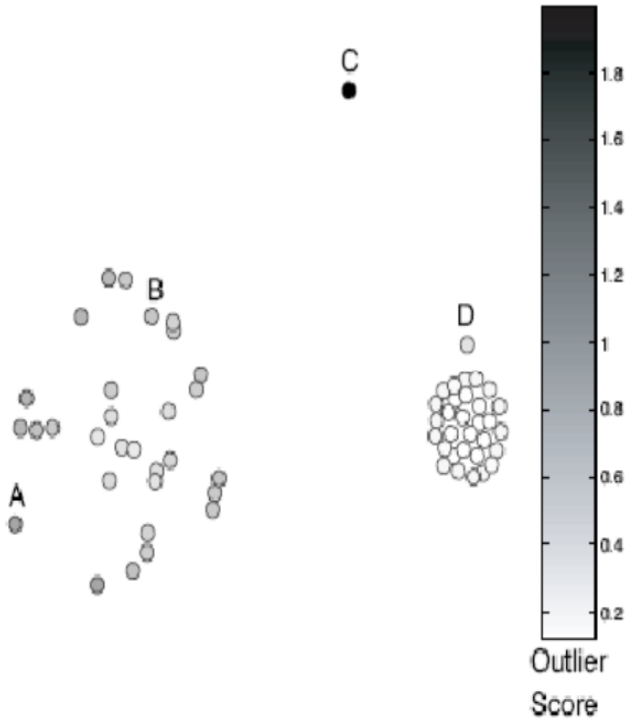


Figure 10.7. Outlier score based on the distance to the fifth nearest neighbor. Clusters of differing density.

Proximity-based outlier detection

- **Pros:**
  - Easier to define a **proximity measure** for a dataset than determine its **statistical distribution**;
  - **Quantitative measure** of degree to which object is **an outlier**;
  - **Deals** naturally with **multiple modes**.
- **Cons:**
  - **O(n^2)** complexity;
  - **Score sensitive** to choose of **k**;
  - Does **not work well** if data has widely **variable density**.

Density-based outlier detection

Outliers are objects in regions of low density.

- Outlier score is **inverse of density** around **object**;
- Scores usually **based on proximities**;
- **Example scores:**
  - Reciprocal of average distance to **k nearest neighbours**:

$$\text{density}(\mathbf{x}, k) = \left(\frac{1}{k} \sum_{\mathbf{y} \in N(\mathbf{x}, k)} \text{distance}(\mathbf{x}, \mathbf{y}) \right)^{-1}$$

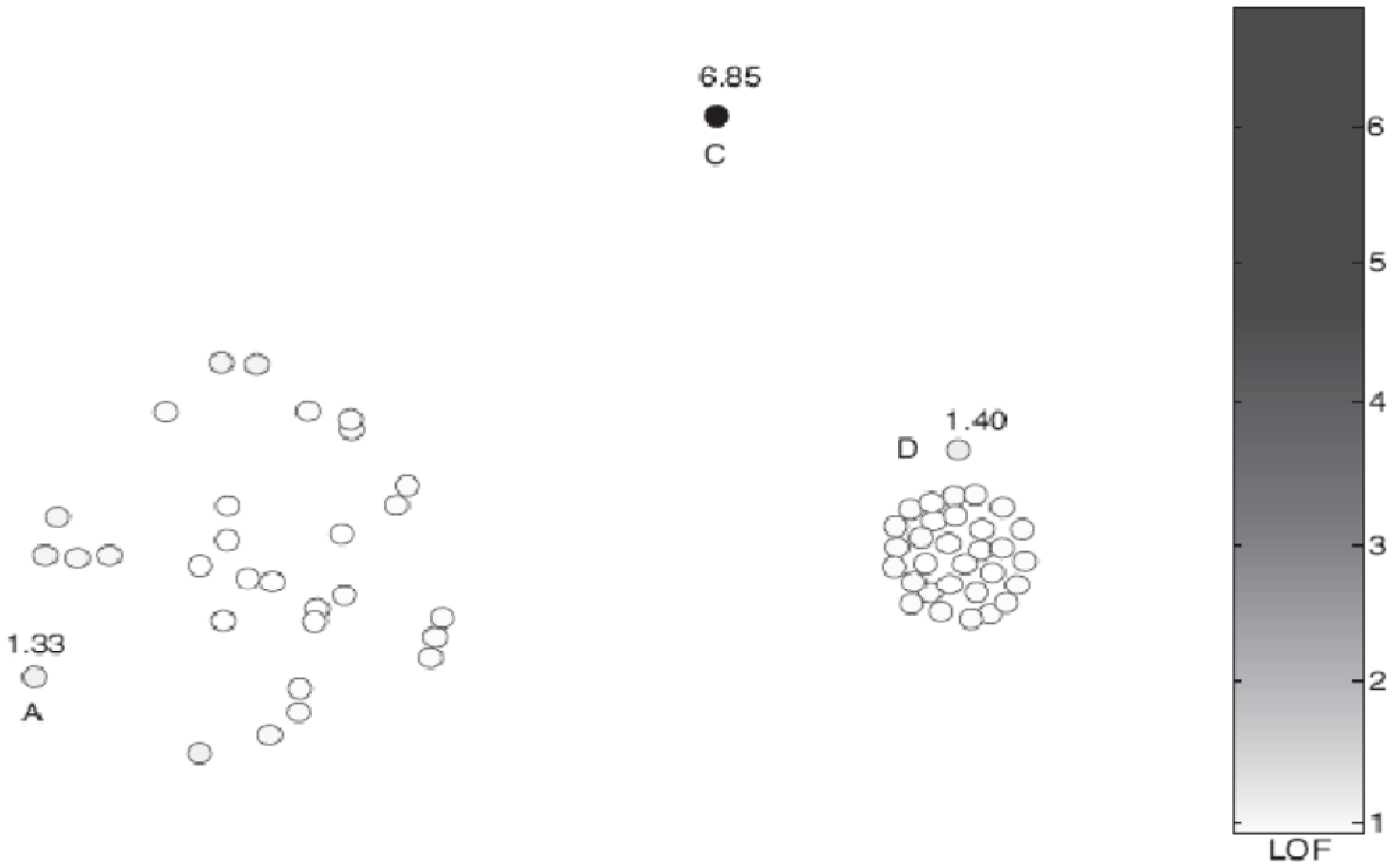
  - Number of objects within **fixed radius d** (DBSCAN);
  - These two example scores **work poorly** if data has **variable density**.

Density-based outlier detection

- **Relative density outlier score** (Local Outlier Factor, LOF):
  - Reciprocal of **average distance to k nearest neighbours**, relative to that of those k neighbours:

$$\text{relative density}(\mathbf{x}, k) = \frac{\text{density}(\mathbf{x}, k)}{\frac{1}{k}\sum_{\mathbf{y} \in N(\mathbf{x},k)} \text{density}(\mathbf{y}, k)}$$

Density-based outlier detection



Density-based outlier detection

- **Pros:**
  - **Quantitative measure** of degree to which object is **an outlier**;
  - Can **work well** even if data has **variable density**.
- **Cons:**
  - **O(n^2)** complexity;
  - Must choose **parameters**:
    - **k** for nearest neighbour;
    - **d** for distance threshold.

Cluster-based outlier detection

Outliers are objects that do not belong strongly to any cluster.

- **Approaches:**
  - **Assess degree** to which **object belongs** to any **cluster**;
  - **Eliminate object(s)** to **improve** objective function;
  - **Discard small** clusters far from **other clusters**.

- **Issue:**
  - **Outliers** may **affect initial** formation of **clusters**.

Cluster-based outlier detection

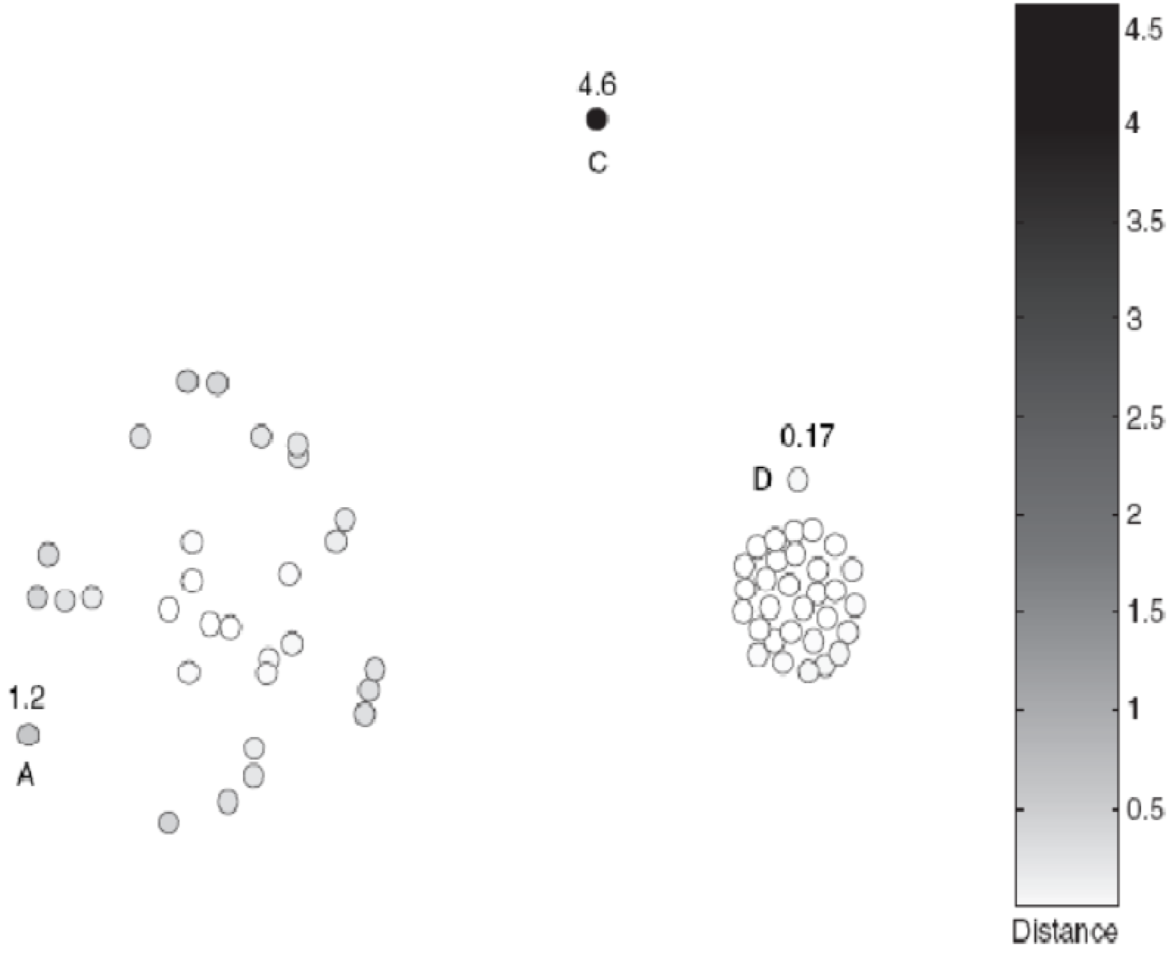
Assess degree to which object belongs to any cluster.

- For **prototype-based clustering** (e.g. k-means), use **distance to cluster centres**.
  - To deal with **variable density clusters**, use **relative distance**:

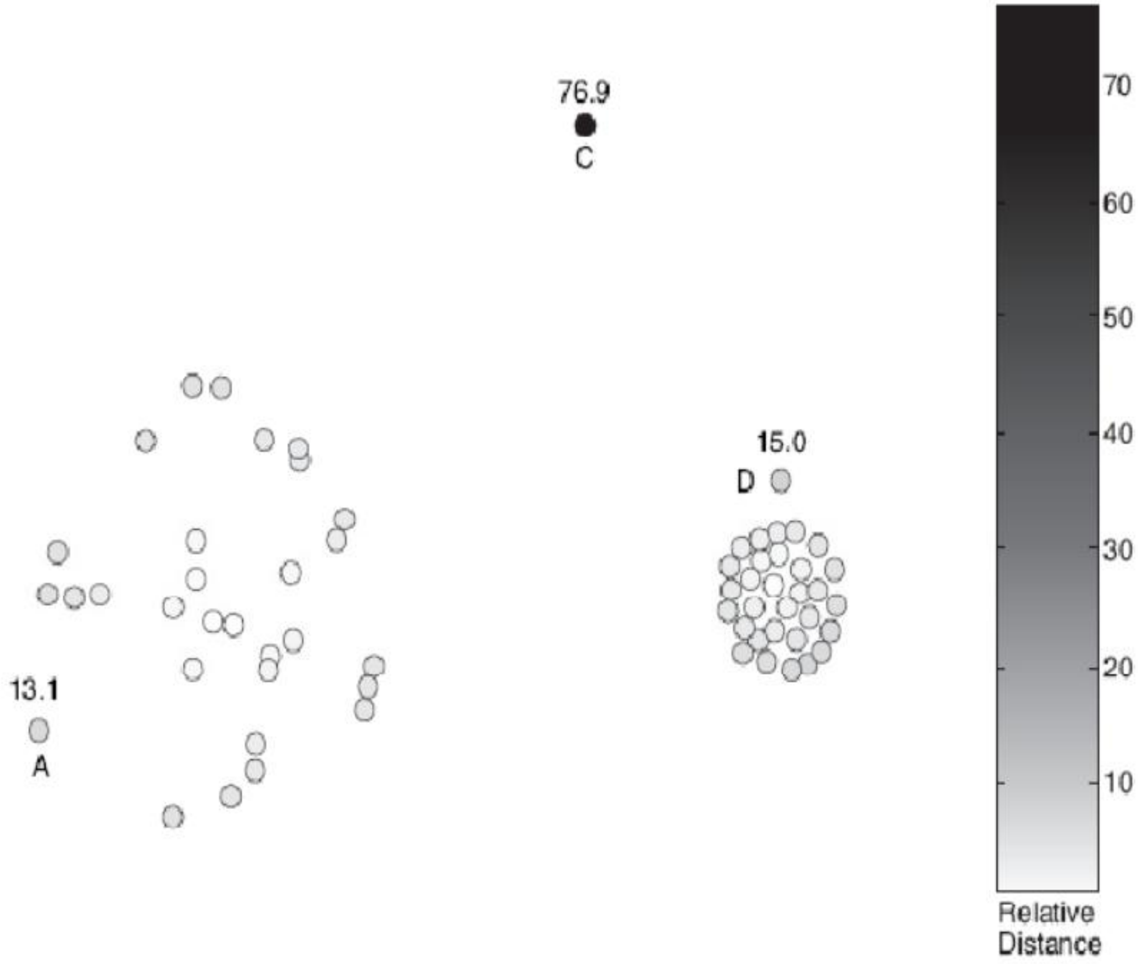
$$\frac{\text{distance}(\mathbf{x}, centroid_C)}{\text{median}(\{\forall_{x' \in C}\ \text{distance}(\mathbf{x}', centroid_C)\})}$$

- Similar concepts for density-based or connectivity-based clusters.

Cluster-based outlier detection



distance of points from nearest centroid



relative distance of points from nearest centroid

Cluster-based outlier detection

➢ **Eliminate object(s) to improve objective function:**

1. Form initial set of clusters;
2. Remove the object which most improves objective function;
3. Repeat step 2) until …

Discard small clusters far from other clusters.

Need to define thresholds for “small” and “far”.

Cluster-based outlier detection

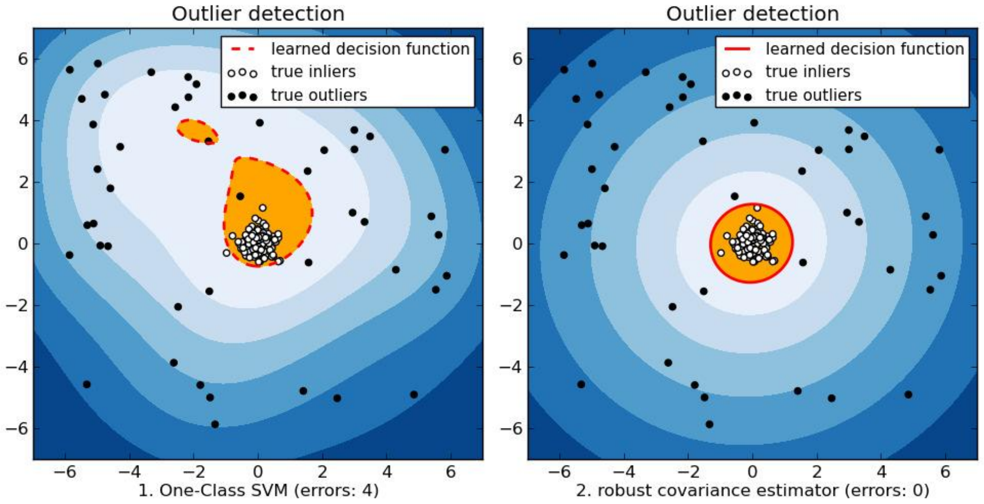
- **Pros:**
  - Some clustering techniques have **O(n) complexity**;
  - **Extends concept** of **outlier** from **single objects** to **groups of objects**.
- **Cons:**
  - Requires **thresholds** for **minimum size** and **distance**;
  - **Sensitive** to number of **clusters chosen**;
  - **Hard to associate** outlier score with **objects**;
  - **Outliers** may affect **initial formation** of clusters.

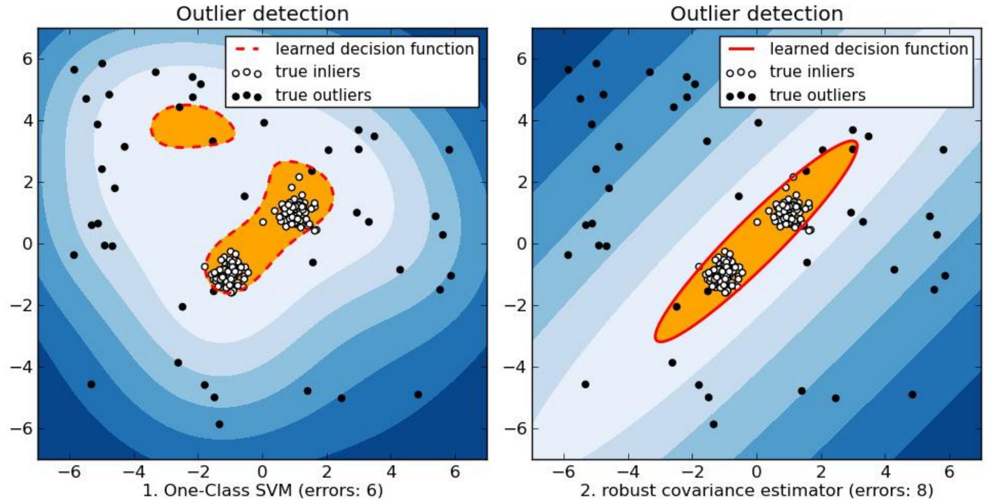
One-class support vector machines

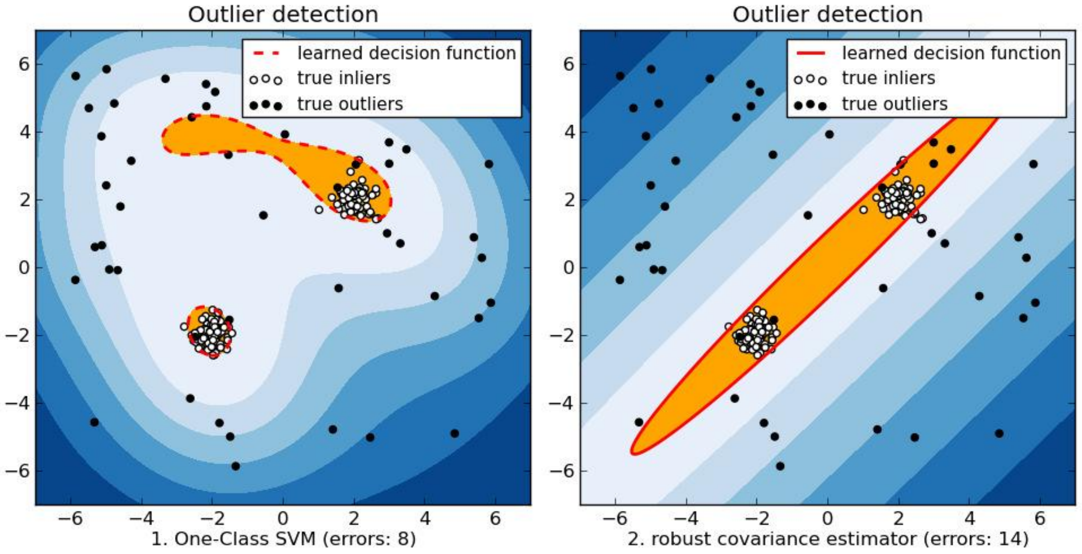
- Data is **unlabelled**, unlike usual **SVM setting**;

- **Goal:**
  - **Find hyperplane** (in higher-dimensional kernel space) which **encloses** as much data as possible with **minimum volume**:
    - Trade-off between **amount of data enclosed** and **tightness of enclosure**, controlled by regularization of slack variables.

One-class SVM vs. Gaussian envelope







Images from:
http://scikitlearn.org/stable/modules/outlier_detection.html

Real-world issues in anomaly detection

- Data often streaming, not static:
  - **Credit card transactions**.

- **Anomalies can be bursty**:
  - Network intrusions.

Questions?

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